

Micro-Lending & Peer Credit Risk Assessor

Architectural Justification

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We chose Layered Architecture for the Micro-Lending & Peer Credit Risk Assessor system.

Architectural Choice

The system is organized into four logical, horizontally-arranged subsystems: Borrower Services (Presentation Layer), Risk & Lending Engine (Business Layer), Financial Services (Service Layer), and Data Layer (Persistence Layer). Each layer communicates only with adjacent layers through well-defined interfaces such as LoanApplication, CreditCheck, LoanApproval, PaymentGateway, NotificationAPI, and UserProfile.

Reason 1: Clear Separation of Concerns

User-facing operations (User Portal, Authentication) remain isolated from credit-risk logic, while payment processing and database management operate independently. This keeps each subsystem focused on a single responsibility and easier to reason about.

Reason 2: Improved Maintainability

Features such as Fraud Detection, Analytics, or the Payment Gateway can be updated or replaced independently without affecting the rest of the application, reducing maintenance effort and improving long-term scalability.

Security Advantage

The layered structure enforces Authentication in the Borrower Services layer, so every loan request must be validated before it can reach the Risk & Lending Engine or Data Layer. Because the User Database and Loan Database sit in an isolated Data Layer that is only reachable through the Loan Manager and Payment Service, borrowers and external clients never interact with the databases directly. This confines sensitive credit-risk and financial data behind controlled interfaces, reducing the attack surface for unauthorized access, data tampering, or SQL injection style attacks.

Performance Benefit

Because layers are decoupled by interfaces rather than direct calls, compute-heavy components like the Credit Risk Engine and Fraud Detection can be scaled, cached, or optimized independently of lightweight components such as the Notification Service or Analytics Dashboard. This prevents a slow, resource-intensive credit check from blocking simple operations like sending a notification, keeping average response times low even as loan-application volume grows.